Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

PENDAHULUAN
TINJAUAN PUSTAKA
beamer
graphicx [latin1]inputenc [T1]fontenc [english]babel listings xcolor
eso-pic mathrsfs url amssymb amsmath multirow hyperref booktabs
tikz
bbm
cooltooltips colordef beamerdefs lvblisting
amssymb amsmath mathrsfs amsthm,amsfonts mathtools
algorithmic [linesnumbered,ruled]algorithm2e float
Author 1 Author 2 Author 3

Title for title page

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Outline

1. Introduction
2. Pre-processing Steps
3. Model Selection
4. Variable Importance and Dimensionality Reduction
5. Results and Conclusion

Formal Problem Setting

- *training set*: inputs $X = (x_1, \ldots, x_n) \in \mathbb{R}^{n \times d}$ and labels $Y = (y_1, \ldots, y_n) \in \mathbb{R}^n$
- *test set*: inputs $X' = (x'_1, \ldots, x'_t) \in \mathbb{R}^{t \times d}$ without labels

Find a function

$$f : X \to Y \tag{1}$$

s.t. the *test set* labels are predicted as accurately as possible, i.e.

$$f(X') \approx Y' \tag{2}$$

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Pre-processing

Several transformations and cleaning steps needed before putting the data into an algorithm, e.g.

Figures/DataPipeline-1.jpg

Gambar: Workflow of Pre-Processing Steps

All transformation need to be preformed on the test set as well!

[basicstyle=,] $\mathbf{basic}_{p}reprocessing = function(X_{c}om, y, scaler =$

$"gaussian")source("replace_{r}atings.R")source("convert_{c}ategoricals.R")source("impute_{d}ata.R")source("encode_{t}ime_{v}ariables.$

dataProcessing

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Optimizing Hyper-parameters

[H] linenosize= i in 1:t Randomly split the data into k folds of the same size
j in 1:k Use jth fold as test set and the union of remaining folds as training set
p in 1:grid Fit model on training set using parameter set p

Predict on test set and calculate RMSE p in 1:grid Calculate average RMSE over the

$t \times k$-runs choose p with the lowest RMSE t-time k-fold crossvalidation and

gridSearch

xgbTuning rfTuning svmTuning

Taking on the curse of Dimensionality

Problem:

- many variables (99 after pre-processing)
- small training set ($n = 1460$)
- variables are correlated with each other

Our approaches:

- Variable selection through variable importance ranking
- Extract a smaller set of variable using PCA

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Results

- Gaussian SVR with all variable is the single best model
- PCA did not work well
- Models perform best with the full set of variables as Figure ?? suggested

Inputs	Gaussian SVR	Random Forest	GBM
All Variables	**0.1308**	0.1484	0.1333
Top 30	0.1323	0.1515	0.1436
PCA	0.1607	0.1657	0.1657

Tabel: RMSE of submitted predictions

Github: finalModels

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